College of Engineering Trivandrum

Department of Electronics and Communication Engineering

ECL332 Communication Engineering Laboratory

Experiment No. 7

Simulation of Pulse Shaping for Transmission

Name:	_____	Date:	_____
Roll No.:	_____	Batch:	_____

1. Aim

1. To transmit an image by mapping the image bits to **BPSK symbols** and convolving with a **pulse shaping filter**.
2. To model a channel for transmission and add **AWGN noise** to the transmitted signal.
3. To reconstruct the image at the receiver by passing through a **matched filter** and demapping the symbols.

2. Learning Outcomes

After completing this experiment, the student will be able to:

- Explain why pulse shaping is required in bandlimited systems (spectral containment and ISI reduction).
- Generate a square-root raised cosine (SRRC) pulse and use it for transmitter pulse shaping.
- Implement upsampling (L) and interpolation via convolution with $p[n]$.
- Add AWGN for a given SNR and simulate a baseband channel.
- Apply matched filtering, compensate for filter delay, downsample correctly, and demap BPSK.
- Reconstruct the transmitted image and compute the bit error rate (BER).

3. Software / Tools Required

- Python (Jupyter / Colab / local Python)
- Packages: `numpy`, `matplotlib`, and an image-reading package such as `opencv-python` (`cv2`) or `imageio`

4. Theory

4.1 Why pulse shaping? (ISI and bandwidth)

In a practical band-limited channel, sharp transitions in the transmitted waveform occupy large bandwidth. If the signal spectrum is "stuffed" into a limited channel without shaping, the channel truncates the spectrum. In the time domain, adjacent pulses overlap, causing **inter-symbol interference (ISI)**, which increases detection errors.

4.2 Sampling viewpoint of ISI

Let the received filter output be sampled at $t_i = iT_b$. A general sampled output can be expressed as

$$y(t_i) = \mu \sum_{k=-\infty}^{\infty} a_k\, p(iT_b - kT_b) = \mu a_i + \mu \sum_{k \neq i} a_k\, p(iT_b - kT_b).$$

For an ISI-free system, the interference term should be zero at sampling instants.

4.3 Matched filtering

At the receiver, optimal detection (in AWGN) uses a **matched filter** whose impulse response is matched to the transmitter pulse. In discrete time, if the pulse shaping filter has impulse response $p[n]$, then the matched filter is

$$g[n] = p[-n].$$

For symmetric SRRC pulses, $p[-n] = p[n]$, so the same SRRC pulse can be used at the receiver.

4.4 Square-root raised cosine (SRRC) pulse

A common pulse shaping function is the SRRC pulse, controlled by:

- **Roll-off factor** β (controls excess bandwidth)
- **Symbol period** T_{sym}

Square-root raised cosine (SRRC) impulse response:

$$p(t) = \frac{1}{\sqrt{T_{\text{sym}}}} \begin{cases} 1 + \beta\left(\frac{4}{\pi} - 1\right), & t = 0, \\ \frac{\beta}{\sqrt{2}}\left[\left(1 + \frac{2}{\pi}\right)\sin\left(\frac{\pi}{4\beta}\right) + \left(1 - \frac{2}{\pi}\right)\cos\left(\frac{\pi}{4\beta}\right)\right], & t = \pm\frac{T_{\text{sym}}}{4\beta}, \\ \frac{4\beta}{\pi}\,\frac{\cos\left((1+\beta)\pi\frac{t}{T_{\text{sym}}}\right) + \frac{T_{\text{sym}}}{4\beta t}\sin\left((1-\beta)\pi\frac{t}{T_{\text{sym}}}\right)}{1 - \left(4\beta\frac{t}{T_{\text{sym}}}\right)^2}, & \text{otherwise.} \end{cases}$$

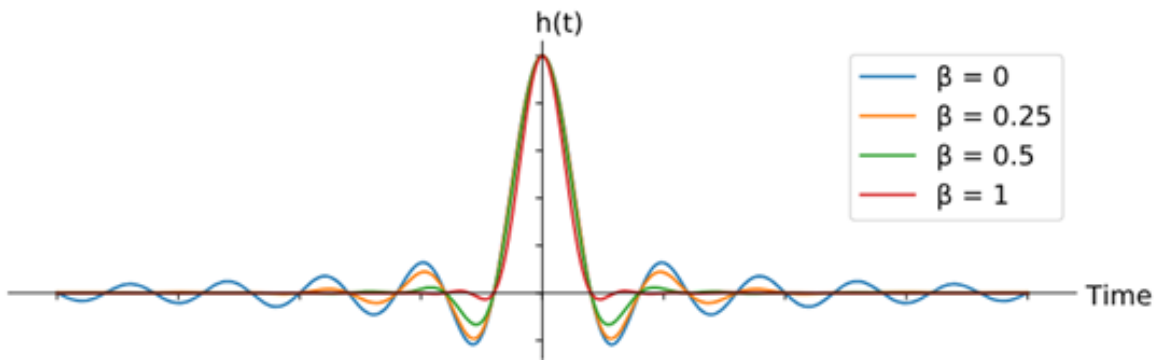


Figure 1: Impulse response of SRRC filter for different values of β.

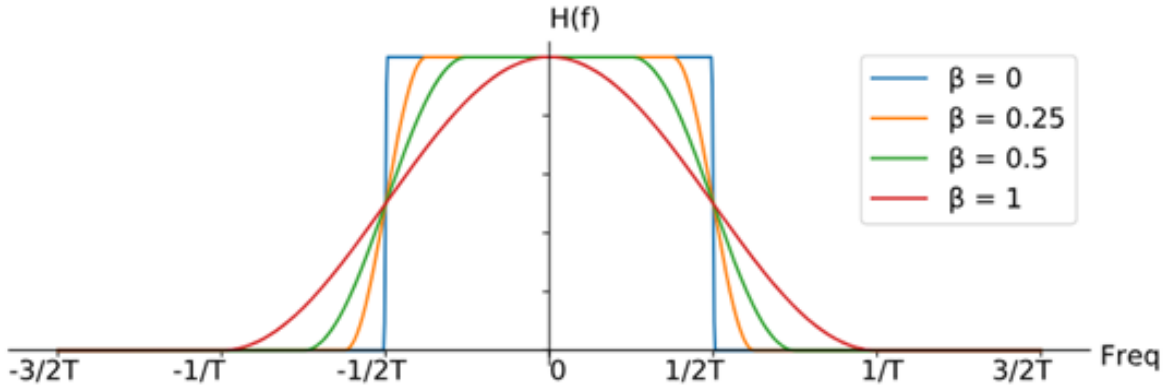
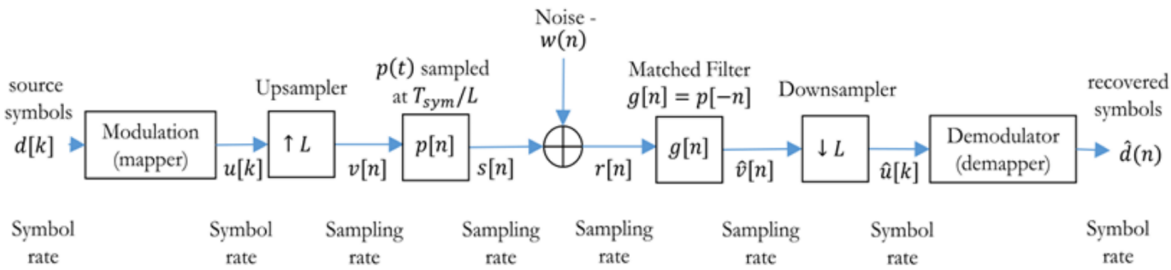


Figure 2: Frequency response of SRRC filter for different values of β.

4.5 System overview

The overall simulation chain is:



5. Experiment Parameters

Use the following default values (can be changed for analysis):

- Upsampling factor: $L = 4$
- Symbol time: $T_{\text{sym}} = 1$
- Pulse span in symbols: $N_{\text{sym}} = 8$
- Roll-off factor: $\beta = 0.5$ (suggested)

Generate time samples for the pulse using:

$$t = -\frac{N_{\text{sym}}}{2} : \frac{1}{L} : \frac{N_{\text{sym}}}{2}.$$

Then compute $p[n] = p(t)$ and use it for convolution.

6. Pre-Lab Questions

1. What is ISI? List two causes of ISI in practical channels.

2. Explain why pulse shaping is important in band-limited communication.

3. For $L = 4$ and $N_{\text{sym}} = 8$, how many samples are in $t = -N_{\text{sym}}/2 : 1/L : N_{\text{sym}}/2$?

4. If the pulse shaping filter length is N, what is the delay (δ) in this linear-phase FIR filter?

5. Why do we apply a matched filter at the receiver in AWGN channels?

7. Procedure

1. **Read image (Input):** Load `cameraman.png` (size 256×256) into Python as a grayscale image array $I[r, c]$.
2. **Flatten:** Convert the 2D image into a 1D sequence

$$x[k] = \text{flatten}(I), \quad k = 0, 1, \ldots, (256 \cdot 256 - 1).$$

3. **8-bit PCM quantization and bit-stream formation:** Since the image is grayscale, perform **8-bit PCM** representation by converting each pixel sample to an 8-bit unsigned binary word. For each sample $x[k] \in \{0, 1, \ldots, 255\}$, generate

$$b_k = \text{bin8}(x[k]) \in \{0, 1\}^8$$

and concatenate all words to form the transmit bit-stream

$$\mathbf{b} = [b_0 \; b_1 \; \cdots \; b_{N-1}],$$

where $N = 256 \cdot 256$ pixels and total bits $= 8N$.
4. **BPSK mapping:** Map each bit to a BPSK symbol sequence $\{a_i\}$ using

$$0 \mapsto -1, \qquad 1 \mapsto +1.$$

Plot the constellation (optional) to verify mapping.

5. **Upsampling by factor L:** Create an impulse-train sequence $u[n]$ by inserting $L-1$ zeros between symbols:

$$u[n] = \begin{cases} a_i, & n = iL \\ 0, & \text{otherwise.} \end{cases}$$

6. **Generate SRRC pulse $p[n]$:** Using parameters L, T_{sym}, N_{sym} and roll-off β, generate the SRRC pulse samples on

$$t = -\frac{N_{\text{sym}}}{2} : \frac{1}{L} : \frac{N_{\text{sym}}}{2},$$

and compute $p[n] = p(t_n)$ using the **square-root raised cosine (SRRC)** impulse response:

$$p(t) = \frac{1}{\sqrt{T_{\text{sym}}}} \begin{cases} 1 + \beta\left(\frac{4}{\pi} - 1\right), & t = 0, \\ \frac{\beta}{\sqrt{2}}\left[\left(1 + \frac{2}{\pi}\right)\sin\left(\frac{\pi}{4\beta}\right) + \left(1 - \frac{2}{\pi}\right)\cos\left(\frac{\pi}{4\beta}\right)\right], & t = \pm\frac{T_{\text{sym}}}{4\beta}, \\ \frac{4\beta}{\pi}\,\frac{\cos\left((1+\beta)\pi\frac{t}{T_{\text{sym}}}\right) + \frac{T_{\text{sym}}}{4\beta t}\sin\left((1-\beta)\pi\frac{t}{T_{\text{sym}}}\right)}{1 - \left(4\beta\frac{t}{T_{\text{sym}}}\right)^2}, & \text{otherwise.} \end{cases}$$

Plot $p[n]$ to verify symmetry and time support.

7. **Transmit pulse shaping (convolution):** Obtain the shaped transmit signal

$$s[n] = u[n] * p[n].$$

Plot a short segment of $s[n]$ to visualize pulse shaping.

8. **Add AWGN (channel) with SNR sweep 0 to 15 dB:** Run the simulation for

$$\text{SNR}_{\text{dB}} = 0, 2, 4, \ldots, 16$$

and compute BER (and reconstructed image) for each SNR.

How to set noise power for a given SNR: Let $s[n]$ be the shaped transmit signal (after convolution). First compute the **average signal power**

$$P_s = \frac{1}{N}\sum_{n=0}^{N-1} |s[n]|^2.$$

Convert the target SNR from dB to linear:

$$\text{SNR}_{\text{lin}} = 10^{\text{SNR}_{\text{dB}}/10}.$$

Then the required **noise power** is

$$P_n = \frac{P_s}{\text{SNR}_{\text{lin}}}.$$

Generate AWGN samples:

- **If $s[n]$ is real-valued:** generate $w[n] \sim \mathcal{N}(0, \sigma^2)$ with

$$\sigma^2 = P_n, \qquad w[n] = \sqrt{P_n}\,\eta[n], \quad \eta[n] \sim \mathcal{N}(0,1).$$

- **If $s[n]$ is complex-valued (I/Q baseband):** set

$$w[n] = \sqrt{\frac{P_n}{2}}\left(\eta_I[n] + j\,\eta_Q[n]\right),$$

where $\eta_I[n], \eta_Q[n] \sim \mathcal{N}(0,1)$ are independent. (This ensures total noise power $E\{|w[n]|^2\} = P_n$.)

Finally form the received sequence:

$$r[n] = s[n] + w[n].$$

9. **Matched filtering:** Form matched filter

$$g[n] = p[-n],$$

and compute receiver filter output

$$y[n] = r[n] * g[n].$$

(For symmetric SRRC, you may use $g[n] = p[n]$.)

10. **Delay compensation (symbol timing alignment):** If the pulse length is N, single-filter delay is

$$\delta = \frac{N-1}{2},$$

and total delay (Tx+Rx) is 2δ. Discard the first 2δ samples so that symbol sampling starts at index $(2\delta + 1)$.

11. **Downsample (symbol decisions):** Take every L-th sample starting from the first valid sample to obtain symbol estimates:

$$\hat{a}_i = y[2\delta + 1 + iL].$$

12. **BPSK demapping (bit decisions):** Apply the ML decision rule (threshold at 0):

$$\hat{b}_i = \begin{cases} 1, & \hat{a}_i \geq 0 \\ 0, & \hat{a}_i < 0. \end{cases}$$

13. **Rebuild 8-bit PCM samples:** Group the detected bit-stream into 8-bit words:

$$\hat{b}_k \in \{0,1\}^8, \quad k = 0, \ldots, N-1,$$

convert each word back to an integer $\hat{x}[k] \in [0, 255]$.

14. **Reconstruct the image:** Reshape the recovered sample sequence into 256×256:

$$\hat{I} = \text{reshape}(\hat{x}, 256, 256),$$

and display the reconstructed image.

15. **Compute BER:** Compare transmitted and detected bits and compute

$$\text{BER} = \frac{\#\{\text{bit errors}\}}{\text{total bits}}.$$

8. Observation Tables

8.1 Parameter values used

Parameter	**Value**
Image size	
Bits per pixel used	
Total bits transmitted	
Modulation	BPSK

Parameter	**Value**
L (upsampling factor)	
T_{sym}	
N_{sym}	
β (roll-off)	

8.2 Filter delay bookkeeping

Quantity	**Expression**	**Value**
Pulse filter length	N	
Single filter delay	$\delta = \frac{N-1}{2}$	
Total delay (Tx+Rx)	2δ	
First valid sample index	$2\delta + 1$	

9. Outputs (Draw Figures)

Insert the following plots after running the code.

- **BPSK constellation plot**

- **SRRC impulse response $p[n]$**

- **Upsampled symbol sequence (few samples)**

- **Transmitted shaped signal (few samples)**

- **BER output / BER vs SNR plot**

10. Result

(List the results you obtained from this simulation)

11. Inference / Analysis

(List your inferences/analysis from this simulation. What have you understood from the experiment and from the plots?)

12. Post-Lab Questions

1. What is the purpose of **upsampling by a factor** L in your code? What would happen if you skip the upsampling step and directly convolve the BPSK symbols with the SRRC pulse?

2. From your SRRC plot, how did the pulse shape change when you varied β?

3. Why did you apply a **matched filter** at the receiver? In your implementation, was $g[n] = p[n]$ or $g[n] = p[-n]$? Why?

4. How did you compute the **total delay** introduced by transmitter and receiver filtering? What starting index did you use for correct downsampling?

5. Explain how you performed **downsampling** (which sample indices were taken). What error did you observe (in bits/image) if you downsample from a wrong index?

Student's Signature: _____